William Horn
Lab 3

1) Change Camera Coordinate Accuracy and Redo Bundle Block Adjustment (6 points)

- Provide a table comparing the total camera position error and total marker position error of both the reference run and modified run.

Accuracy	(C 10m 10˚, M 0.1m)	(C 0.1m 1˚, 10m)
Camera Error (m)	**5.811**	**0.179**
Markers Error (m)	**0.083**	**3.938**

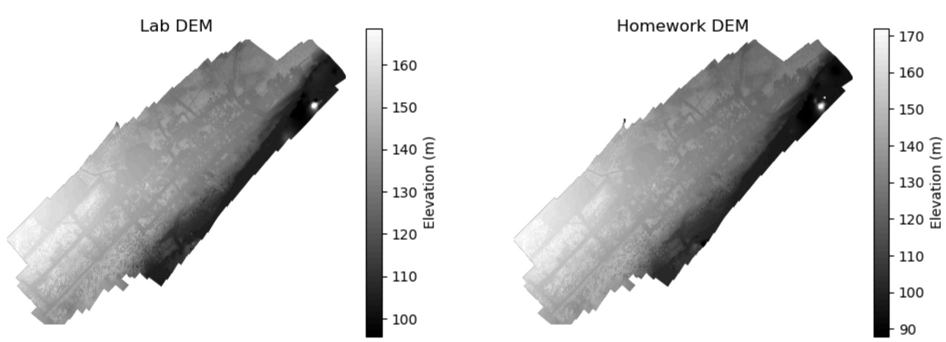
- Provide a summary description of how these error numbers have changed

Looks like they flipped. Camera error reduced, marker error increased

- Provide an explanation of the observed changes

A higher accuracy means a smaller error, lower accuracy means larger error. If we are more confident in where the camera is, we have less error our images, but because we are less certain of our ground control points they end up having more error.

3) Export DEM of Modified Run and Export Modified Project Report



4) Compare DEMs of Reference and Modified Run

- Provide a summary of the DEM differences that you can identify.

DEM Difference (Lab - Homework) mean: 2.91, std: 3.98

Overall they are pretty close. The lab DEM seems to be more uniform than the homework dem. The water in the homework dem is lower by about 10 meters. The area by the clearing is about 5 meters higher in the homework DEM as well.

- Provide an explanation for why the DEMs might be different in the observed way

Changing the accuracy could have made the model not fit as well so the whole dem ends up being tilted. This is why the clearing is higher and the water is lower.

5) Compare Reference and Modified Run Project Reports

- Provide a summary of the differences that you can see in the plots of camera distortions?

It's hard to tell. I don't see much difference between the two plots. The homework plot looks like it has a bit more red lines.

- Provide a summary of the difference that you see in the estimated camera distortion parameters?

The majority of the numbers in the coefficient and correlation matrix are different.

- Provide an explanation for the observed changes?

It's already hard to tell what is internal to the camera and what is from the image. The different accuracies changed the calculated camera calibration.

- Comment on how the absolute error and the distribution of errors of camera positions change?

In the lab report all the ellipses are aligned in the same direction and have a uniform gradient in X/Y and Z error that increases the closer to the water the camera is. The homework ellipses are all over the place. They have error outlets but most of the X/Y/Z error is lower.

- Explain the changes that you are seeing

If we *think* are certain of where the camera is, it can lead to more error because relating all of the images together ends up being much harder. It's more important to be certain of your ground control points than your camera locations.